Christopher Kuizon

I'm a machine learning engineer. I spend most of my time on the part after training: checking whether a model still works on messy, real inputs.

ML Engineer at TrueCar. CS and Business at USC, continuing into an M.S. in CS. Based in Los Angeles.

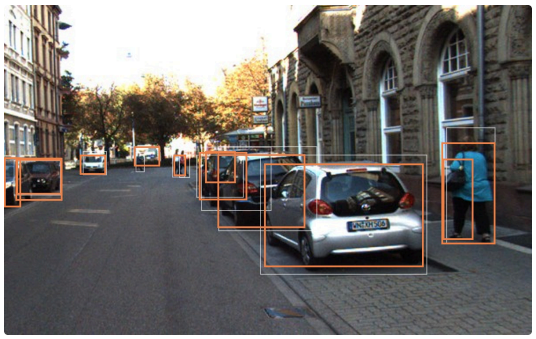
Email christopher.kuizon@gmail.com LinkedIn linkedin.com/in/christopherkuizon

GitHub github.com/Khriis-K

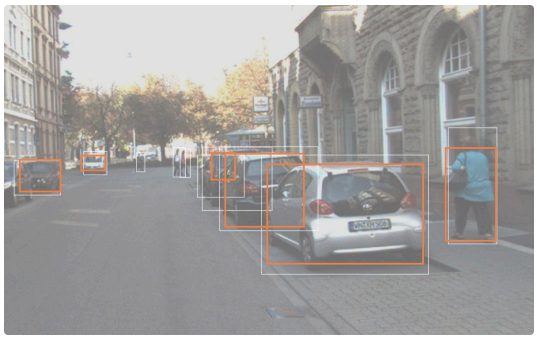
INTERACTIVE VERSION

khriis-k.github.io

Live demos, video, and the fog slider



clear · 12 detections ≥ 0.25



fog severity 0.60 · 6 detections ≥ 0.25

Same frame, same detector, more fog. Orange boxes are detections; thin white outlines are the labeled objects. Fog and detector from the Edge Perception Reliability Lab. On the website this is a slider. Frame: KITTI (Geiger et al., CVPR 2012), cropped, CC BY-NC-SA 3.0.

SELECTED WORK

Tether

Personal project, Oct 2026 to now

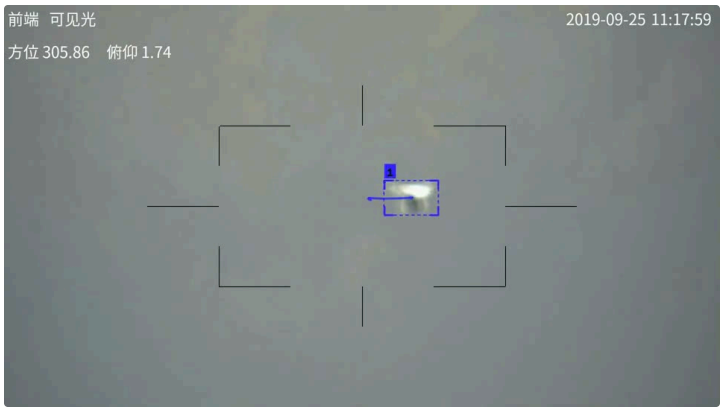
A real-time multi-object tracker in C++ that keeps a steady ID on each person or drone, even when it's briefly hidden or the detector misses it. YOLO26n finds objects in each frame; a hand-written Kalman filter, staged Hungarian matching, and a track lifecycle link them over time.

I prototyped the tracker in Python, then ported it to C++ with identical output on 462 real detection files, and moved it onto the GPU with TensorRT, a custom CUDA kernel, and a parallel pipeline. It runs at 195 FPS on a laptop against a 120 FPS goal, and matches ByteTrack on MOT17. Not every goal was met: holding a drone's ID through dropouts fell 5.3 points short, and the scorecard says so.

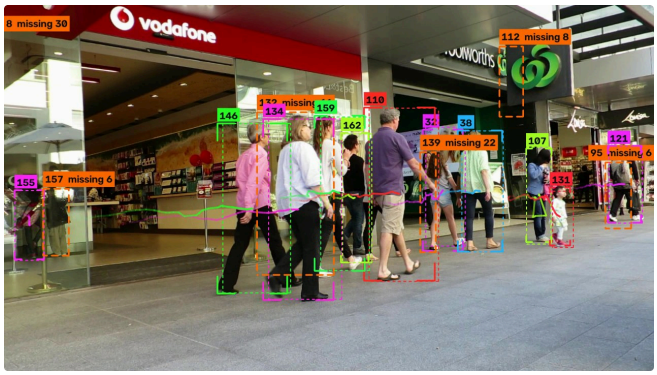
C++, CUDA, TensorRT, Python, YOLO

Project khriis-k.github.io/Tether

Source github.com/Khriis-K/Tether



The fine-tuned drone model on an Anti-UAV test clip. Video: Anti-UAV, MIT License.



Tracking pedestrians on MOT17-09. Dashed orange boxes are people the detector lost, which the tracker is still searching for. Video: MOT17, CC BY-NC-SA 3.0.

Goals set before measuring, on one laptop GPU

120+ FPS on a video file	195 FPS	MET
Worst 1% frame time under 16 ms	6.5 ms	MET
Match ByteTrack on MOT17 (HOTA, IDF1, ID switches)	HOTA 41.5 vs 40.9	MET
C++ tracking step 10×+ faster than Python	14.1×	MET
Keep a drone's ID with 20% of detections deleted	84.0% vs 89.3% goal	MISSED

Edge Perception Reliability Lab

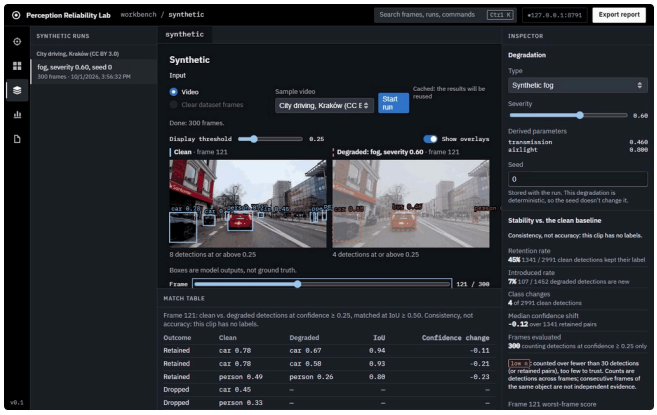
Personal project, Sep 2026 to now

Does a detector trained on clear roads still work in fog, snow, and at night? This lab runs YOLOX-Nano over 2,000 driving frames in 8 weather and lighting conditions, and compares real bad weather against simulated degradation.

On a 300-frame benchmark with 2,946 labeled objects, mAP dropped 85% under severe simulated fog. The scoring engine that classifies hits, misses, false alarms, and class mix-ups has 800+ automated tests.

Python, FastAPI, React, TypeScript

Project	khriis-k.github.io/Edge-Perception-Reliability-Lab
Source	github.com/Khriis-K/Edge-Perception-Reliability-Lab



The workbench during a synthetic fog run on 300 frames of city driving in Kraków.

MyCouncil

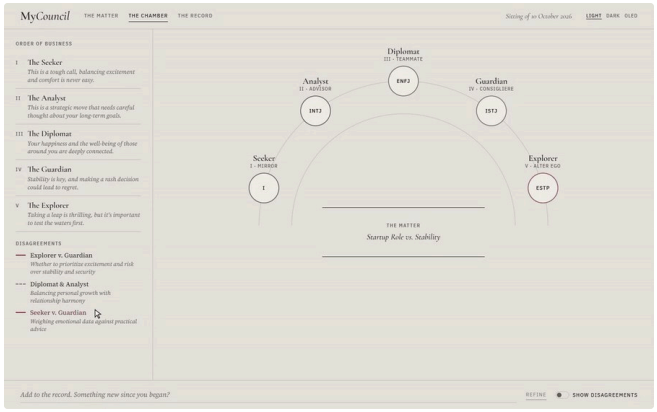
Personal project, Nov 2025 to now

You bring a hard decision and 16 AI counselors with different thinking styles argue it out. You can debate them, refine the dilemma, or talk to one counselor alone.

The hard part was memory. Counselors recall what you said in earlier sessions through a retrieval pipeline with BGE embeddings and a reranker. It improved relevant-context recall by 168% on 160 held-out tests, measured within a 240-case suite that also tracks ranking quality and latency.

Python, Node.js, React, TypeScript

Project	khriis-k.github.io/MyCouncil
Source	github.com/Khriis-K/MyCouncil



A council forming around "move for the job, or stay with friends?"

What caused 20 years of runway incursions

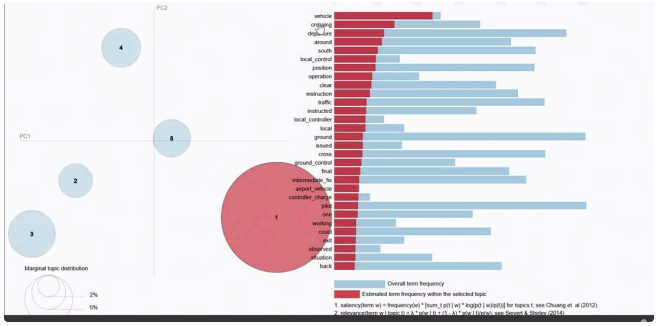
Capstone with the USC Aviation Safety & Security Program, Mar to Jun 2025. Our team of eight students was selected by our professor; I built the topic models.

NASA's Aviation Safety Reporting System holds pilot and controller narratives of runway incursions going back to 2001. We analyzed 11K+ of these reports. Researchers have mostly coded them by hand. We used NLP to find the recurring causes automatically and to see how they changed over time.

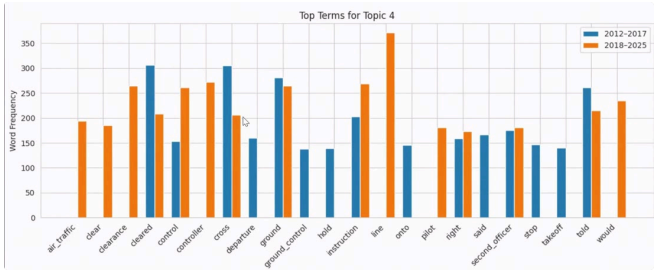
I built the LDA and BERTopic pipelines, including expanding aviation abbreviations in the preprocessing step and tuning UMAP and HDBSCAN so the topics separated cleanly. Then I compared topics before and after 2017, which showed a 105% increase in risk on time-pressure incidents. We presented the findings to 50+ program representatives.

Python, gensim, BERTopic, sentence-transformers, React, FastAPI

Dashboard	khriis-k.github.io/ASRS-Dashboard
Source	github.com/Khriis-K/ASRS-Dashboard



LDA topics for 2012 to 2017. Selecting a topic shows its most relevant terms (red) against overall frequency (blue). Interactive on the dashboard.



One topic, two periods. "Line" and "clearance" appear only after 2018, while "hold" and "takeoff" drop out.

Rebuilding sales attribution at TrueCar

Machine Learning Engineer, May 2026 to now. Proprietary, so no code here.

TrueCar's legacy sales attribution algorithm used a probabilistic ranking built on a naive Bayes model.

I rebuilt the ranking and trained an XGBoost model on 52 million labeled pairs with 10 target-encoded features.

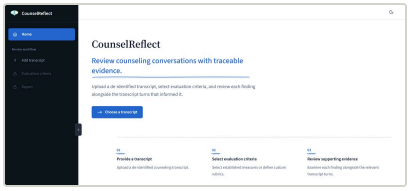
XGBoost, Spark, AWS

Change versus the legacy system, at over 95% confidence

Billable sales in production	+124%
Monthly match volume	+196%
Match accuracy	+19%
Top-1 accuracy vs. naive Bayes	+59%
AUC-ROC	+30%
Mean reciprocal rank	+29%

TrojanSafe Housing A web app that maps historical crime data from USC Department of Public Safety daily incident logs, so students and their families can compare off-campus housing. Private repo, code available on request.

Counsel Reflect A tool for reviewing counseling conversations turn by turn, with 81 emotion, safety, and factuality metrics. I helped design the interface and React dashboard, which scored 78.8 and 75.2 on usability against a 68 benchmark. counselreflect.com



EDUCATION

University of Southern California, Los Angeles — Expected May 2028

B.S. Computer Science and Business Administration. Progressive M.S. in Computer Science, minor in Artificial Intelligence Applications. GPA 3.83.

SKILLS

Languages	Python, C++, Java, Kotlin, SQL, JavaScript, HTML, CSS, PHP
ML	PyTorch, TensorFlow, Keras, Hugging Face, XGBoost, spaCy, NLTK
Tools	CUDA, TensorRT, Docker, Git, Linux, Spark, Airflow, AWS, Slurm, React, FastAPI

EXPERIENCE

Machine Learning Engineer, TrueCar — May 2026 to now

Rebuilt sales attribution ranking, and wrote an Airflow DAG that automates a 7-day manual refresh of its probability tables.

Full-Stack Software Engineer, Counsel Reflect — May 2025 to Aug 2026

Fine-tuned RoBERTa classifiers to 80 to 92% accuracy across three empathy dimensions.

Data Analyst, USC Aviation Safety & Security Program — Mar to Jun 2025

Teaching Assistant, USC machine learning courses — Jan 2025 to now

Four courses, from intro AI to NLP and generative AI: TAC 259, 359, 449, and 459. I've mentored 100+ students across 90+ projects.

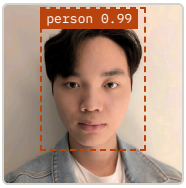
Social Chair, EscapeSC — Fall 2026

USC's escape room club. We design and build rooms on campus.

Co-Chair, Viterbi Undergraduate Student Association — Feb 2024 to Aug 2026

Represented 50+ engineering clubs to the dean and started the school's first AI ethics panel.

ABOUT



I'm most interested in ranking, retrieval, and applied ML that has to hold up in the real world, whether it's serving millions of users or running somewhere mistakes are costly.

I also TA machine learning courses, so if I'm not building something, I'm probably explaining why someone else's model isn't training. Otherwise I'm with EscapeSC, USC's escape room club, where we build rooms on campus. My friends and I are 17 rooms in.

The fastest way to reach me is email: christopher.kuizon@gmail.com **Full site:** khriis-k.github.io